

Generating a Paracosm for Training-Free Zero-Shot Composed Image Retrieval



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What is Composed Image Retrieval (CIR) ?

CIR aims to retrieve an image from a database that matches the user-uploaded query.

multimodal query



is taken from the interior
of a car

target image?



database



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The multimodal query only implicitly defines a “mental image”, which does *not* physically exist!

Generating the "mental image" and a brief description

We generate the "mental image" and its description for the given query using LMMs.

multimodal query



is taken from the interior
of a car

LMM



target image?



database



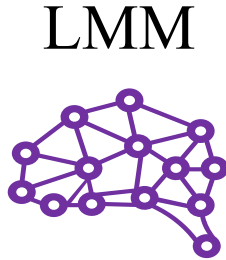
Generating the "mental image" and a brief description

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multimodal query



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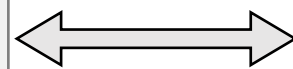


mental image



Two horses standing on a
grassy field viewed from
inside a car.

matching

A double-headed grey arrow indicating a matching process between the mental image and the target image.

target image?



database



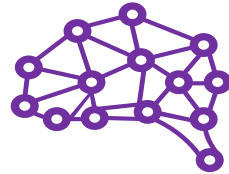
Synthetic-to-real domain gap!

“Mental image” has synthetic-to-real domain gaps compared to the real images in the database.

multimodal query



is taken from the interior
of a car



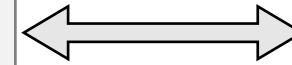
synthetic-to-real domain gaps!

mental image



Two horses standing on a
grassy field viewed from
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matching



target image?



database



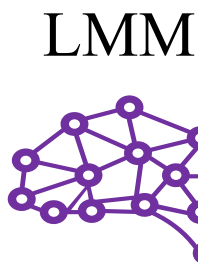
Domain gap mitigation

Generating synthetic counterparts for database images and use them to facilitate matching.

multimodal query



is taken from the interior
of a car

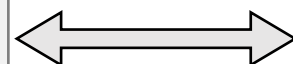


mental image

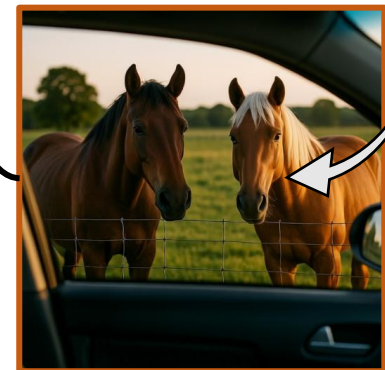
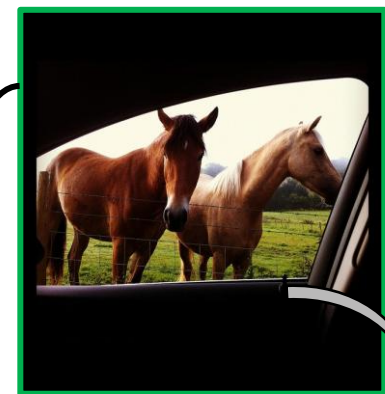


Two horses standing on a
grassy field viewed from
inside a car.

matching



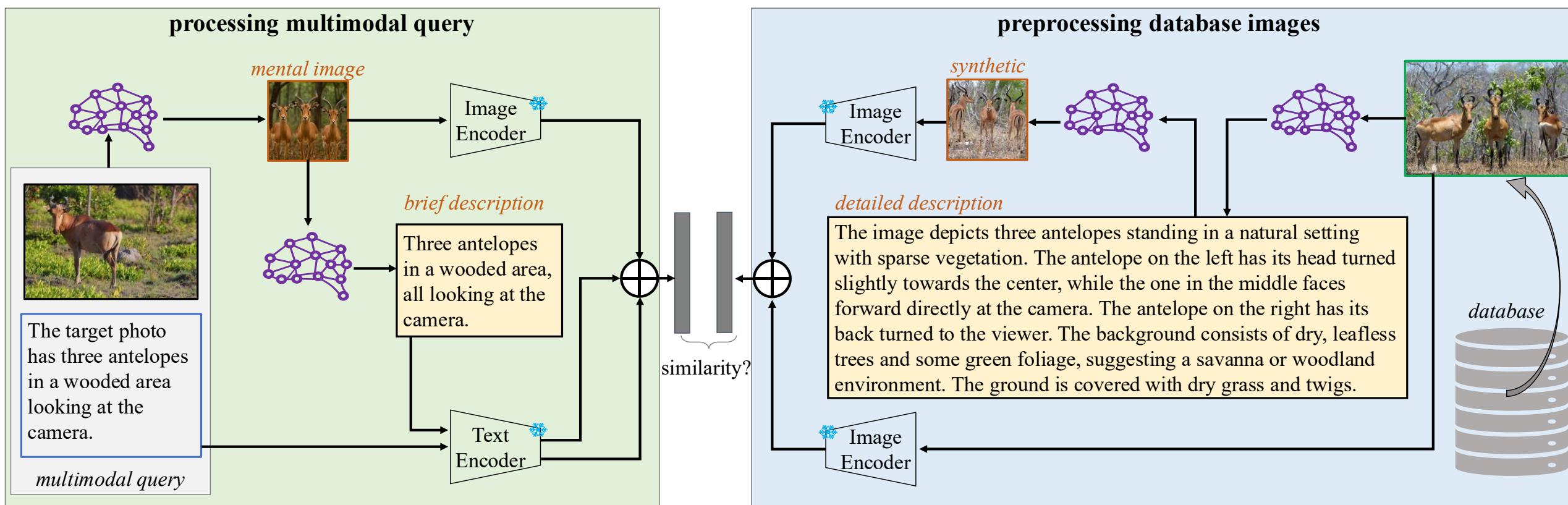
target image?



synthetic counterpart

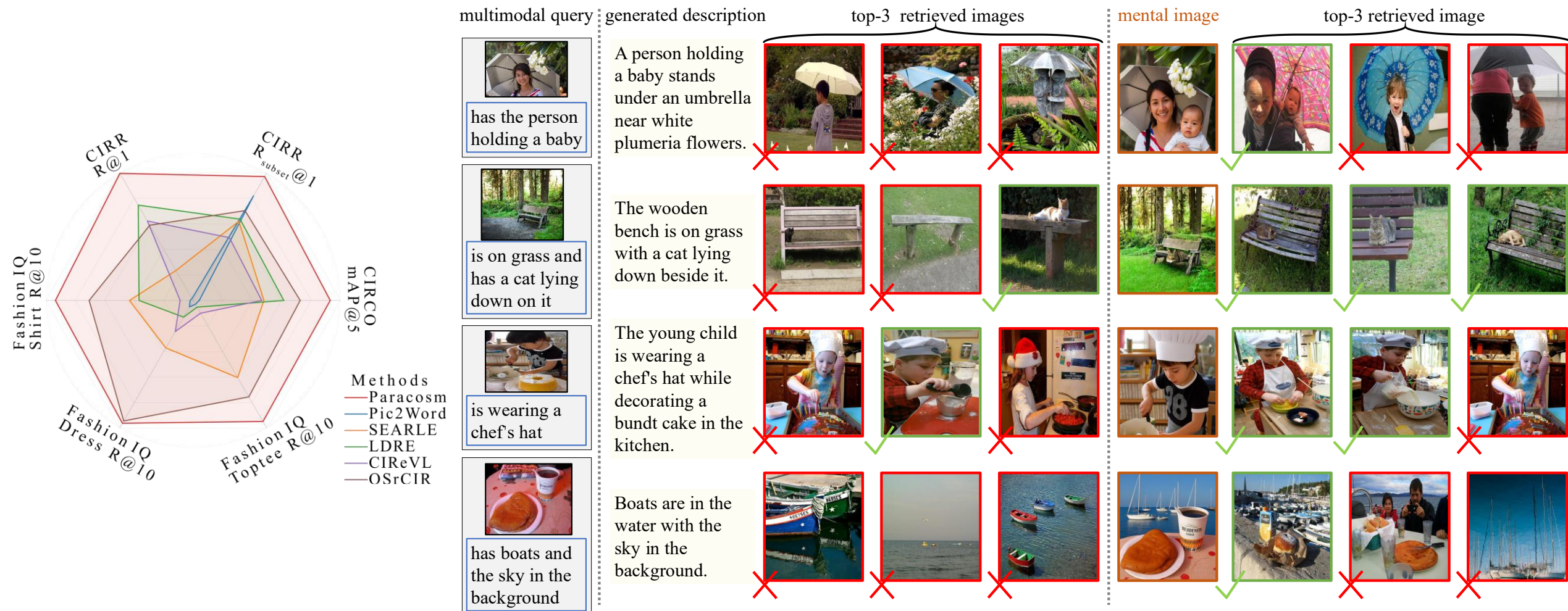
Paracosm

A space including features of real and synthetic images, and texts.



Paracosm

Resoundingly outperforming existing zero-shot CIR methods.



Paracosm

Failing due to limitations of generating implausible and counterfactual mental images.

multimodal query



is of a different color and has a duck next to it

mental image



retrieved image by Paracosm



target image



multimodal query



has a different color and there is a door next to it

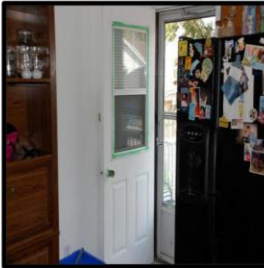
mental image



retrieved image by Paracosm



target image



is open and has gas burners nearby



has the woman and the man with roles switched



Thanks!

Paracosm

- solves CIR from first principles by generating "mental images".
- mitigates the synthetic-to-real domain gap by synthesizing database images.
- significantly outperforms existing ZS-CIR approaches on established benchmarks.



Project Page